

DATA VORTEX A'26

Round 2 — Evaluation Metrics Report

Rebuilding the Social Engine — Semantic Understanding Layer

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1. Evaluation Protocol

Final metrics are calculated on the held-out test partition of 1,817 records. Accuracy, precision, recall and Macro F1 are used, with Macro F1 emphasized for the imbalanced topic task.

Partition	Rows	Unique texts	Text overlap
Training	7,183	6,320	0
Testing	1,817	1,580	0

2. Final Model Summary

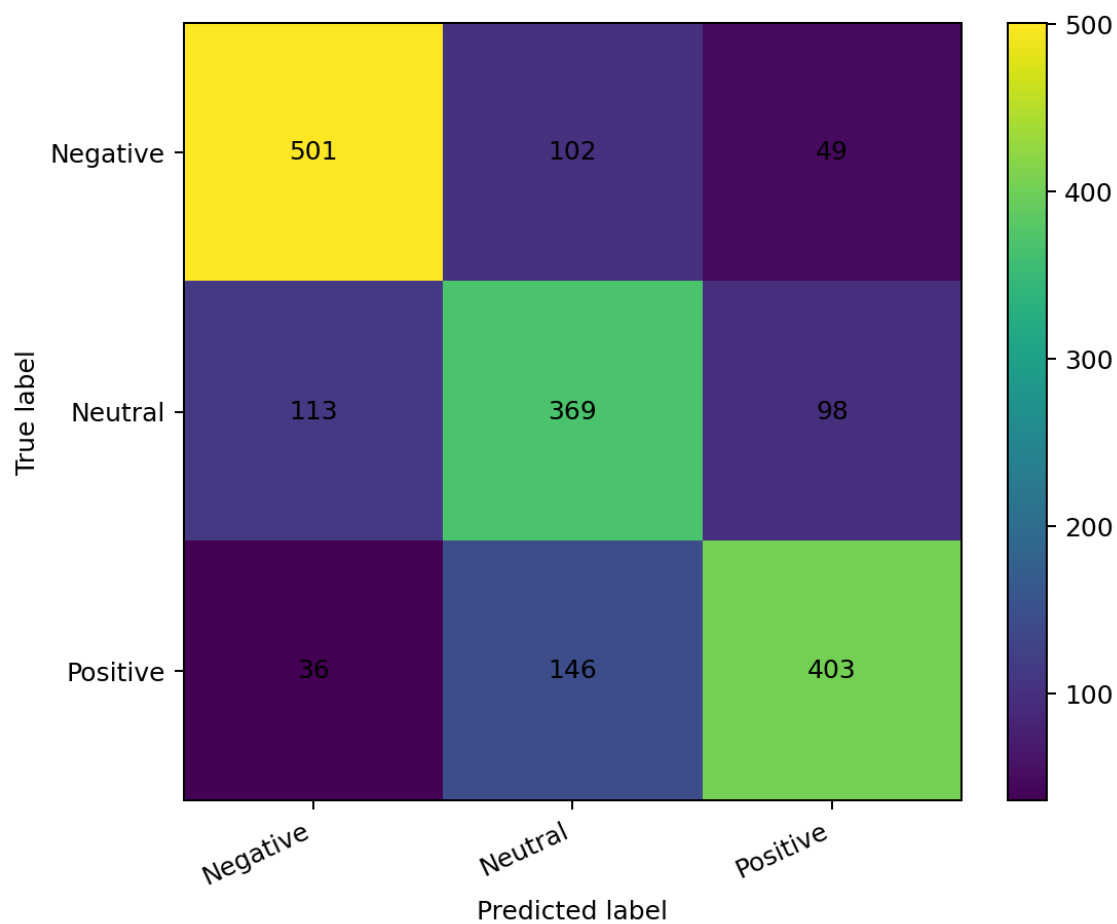
Task	Model	Accuracy	Macro F1
Sentiment	DistilBERT	70.06%	0.6988
Topic	Word + Char TF-IDF + Balanced Logistic Regression	94.22%	0.6939

3. Sentiment — Final Classification Report

Class	Precision	Recall	F1	Support
Negative	0.7708	0.7684	0.7696	652
Neutral	0.5981	0.6362	0.6165	580
Positive	0.7327	0.6889	0.7101	585
Macro Avg	0.7005	0.6978	0.6988	1817
Weighted Avg	0.7034	0.7006	0.7016	1817

Accuracy 70.06%; Macro Precision 0.7005; Macro Recall 0.6978; Macro F1 0.6988; Weighted F1 0.7016.

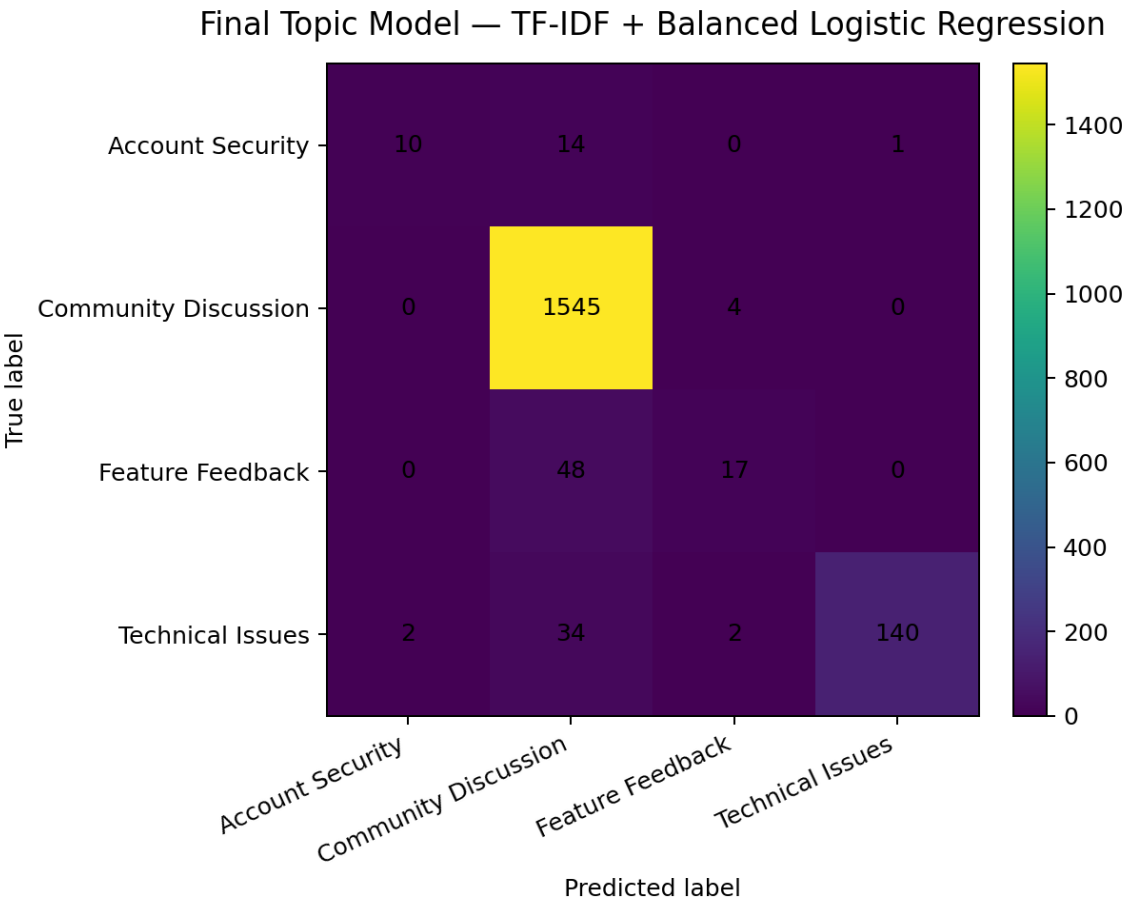
Final Sentiment Model — DistilBERT



4. Topic — Final Classification Report

Class	Precision	Recall	F1	Support
Account_Security	0.83	0.40	0.54	25
Community_Discussion	0.94	1.00	0.97	1549
Feature_Feedback	0.74	0.26	0.39	65
Technical_Issues	1.00	0.79	0.88	178

Accuracy 94.22%; Macro F1 0.6939. Test support: Account_Security 25, Community_Discussion 1,549, Feature_Feedback 65, Technical_Issues 178.



5. Model Comparison Record

Experiment	Task	Accuracy	Macro F1	Status
Combined Word + Char TF-IDF + Balanced LR	Sentiment	61.14%	0.6121	Used as classical baseline
Combined Word + Char TF-IDF + Balanced SVM	Sentiment	60.43%	0.6050	Rejected
Custom class-weighted LR	Topic	91.74%	0.5225	Rejected
Word trigrams + Char TF-IDF	Sentiment	60.98%	0.6107	Rejected
Word trigrams + Char TF-IDF	Topic	94.11%	0.6804	Rejected
Tuned LR, C=2	Sentiment	61.36%	0.6147	Retained before transformer stage
Tuned LR, C=0.25	Topic	93.62%	0.6797	Rejected on held-out Macro F1
ComplementNB	Sentiment	61.59%	0.6098	Rejected
ComplementNB	Topic	86.46%	0.2866	Rejected

6. Error Metrics

Sentiment error transition	Count
Positive → Neutral	146
Neutral → Negative	113
Negative → Neutral	102
Neutral → Positive	98
Negative → Positive	49
Positive → Negative	36
Total sentiment errors	544

Topic error transition	Count
Feature_Feedback → Community_Discussion	48
Technical_Issues → Community_Discussion	34
Account_Security → Community_Discussion	15
Community_Discussion → Feature_Feedback	4
Technical_Issues → Account_Security	2
Technical_Issues → Feature_Feedback	2
Total topic errors	105

7. Interpretation

Sentiment is evaluated on balanced labels, while topic classification is heavily imbalanced. The topic model's 94.22% accuracy should therefore be read alongside Macro F1 0.6939 and the minority-class F1 values: 0.54 for Account_Security and 0.39 for Feature_Feedback.